

ecute the attacks, the defacements are taken more seriously than they should be, and alarmist media blasts video streams with news of the defacement of the usually obscure and specialised web sites. These kinds of attacks have been going on for years, and show no signs of stopping.

It is not just websites that are left unsecured, there are still people stupid enough to actually remove the default passwords on their WiFi devices. This is liable to attract all kinds of strangers nose-deep in their touchscreens to gather in the vicinity of the hotspot. However, it is equally stupid to connect to an unsecured WiFi, as these open hotspots can be used to gather credentials of those who login, and execute the man in the middle attacks outlined above. The best thing to do is secure your own hotspot, as well as not access an unsecure hotspot.

Users are liable to make all kinds of assumptions about security that is simply not true. One of the most common assumptions is that Apple devices and Linux machines are invulnerable to malware attacks. While generally, these systems are more secure as compared to say Android smartphones and Windows machines, there are a number of reasons. Many kinds of malware attacks have been taking place on these systems, and as the adoption of Apple devices and Linux machines becomes more mainstream, the instances of malware attacks can only increase. Expect more Linux and Apple malware in 2018.

Another stupid assumption is the extent to which the private modes such as the Incognito mode in Chrome and Private Browsing modes in Opera and Firefox mask the activity of the user. Chrome says it upfront that the Incognito mode will not hide the browsing activity from your employer, your ISP and the websites you visit. Websites you visit can still track your movements across the web using advanced and persistent cookies, along with your IP address. The private browsing modes simply do not keep a local copy of the websites you visited, on the computer. To be truly anonymous while browsing the web, use a good VPN service. ■

MASS THREATS

Not all tech threats in 2018 will affect you directly. Some will target other things - organisations and services - which will affect your life. Some will target a large group of people, of which you could be a part. Undoubtedly, you also need to be prepared for these.

It's not just your life that is getting increasingly connected to the internet - it is also the infrastructure around you. Everything from transportation systems to power grids to even missile defence systems have some connection to the internet. And while that might make them all the better and oh-so-sleek, it does also leave them open to attacks.



Credit: US Department of Defense

The US military has already tested out swarms of these autonomous Perdix micro drones, with the potential for weaponisation, and deployed from an aeroplane. The concept is explored in the speculative short film, *Slaughterbot*.

One of the first things that you need to worry about are drones. Expect this year to be the year when you hear about the first drone terrorist attacks. We're not talking about huge, noisy drones here. Even something as small as the Mavic Pro can be fitted with a payload delivery system that can carry upto 400g of C4, which can be detonated from a reasonable distance to cause significant damage. And if you think that

time is far-far away, let us tell you that the first drone attack of the year that did not come from a government in power has already happened.

In 2016, an attack on Ukraine's power system plunged 20 percent of Kiev, the capital, into darkness. As we move towards smart cities and integrate more of our utilities to a central control system, applications of attack methods like botnets and social engineering could prove effective in plunging entire cities into darkness, or worse, lockdowns and chaos in coordination with a planned terrorist attack. Think Bane's plan from *Dark Knight Rises*. Even without something that grand, finding loopholes in firmware and software running on older models of aeroplanes, ships and trains could prove to be effective for attackers looking to cause large scale damage. We could be looking at hijackings where the hijackers don't have to be anywhere near the target vehicle.

Another area that will affect you indirectly is gene-related technological advancements. For instance, genetically targeted poisons are very close to achieving species-specific targeting capabilities. If that comes to fruition, we might be able to eradicate entire species of pests and pathogens off the face of the planet - but as it has been widely discussed, we might end up suffering the consequences of playing around with biodiversity.

Not much can be done about these mass threats at an individual level, rather than calling for the right measures and policies through your elected representatives and generating as much awareness as possible. You could also leave the world of tech behind you and live off the land in a remote cabin in the woods. This way, you would no longer need to be terrified of technology. **1**